FEDERAL SPECIFICATION

WAX, CASTING, DENTAL

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope.—This specification covers sheet casting wax and ready made wax shapes suitable for use in dental casting procedures.

1.2 Classification.

1.2.1 *Type, grade, and classes.* — Casting wax covered by this specification shall be of one type, grade, and of the following classes, as specified in the invitation for bids:

Class A.—28-gage sheet.
Class B.—30-gage sheet.
Class C.—Ready made shapes and sizes as shown in 3.4.3.

2. APPLICABLE SPECIFICATIONS

2.1 There are no other specifications applicable to this specification except those referenced in section 7 by individual departments.

(Copies of Federal specifications and the Federal Specifications Index may be obtained upon application, accompanied by check, money order, cash, or Government Printing Office coupons, to the General Services Administration, Business Service Center, Region 3, Seventh and D Streets, S. W., Washington 25, D. C. This office will also honor deposit account numbers issued by the Government Printing Office. Single copies of product specifications required for bidding purposes are available without charge at the GSA Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C. Prices of additional copies may be obtained from the GSA Regional Offices.)

3. REQUIREMENTS

3.1 Material.—Casting wax shall be essentially a mixture of waxes, resins, and hydrocarbons of the paraffin series, combined in such proportions that the resulting material will possess the properties as hereinafter specified.

3.2 Properties.

3.2.1 *Working qualities.*—The wax shall be pliable and readily adaptable at temperatures of 40° to 45° C., and shall copy accurately the surface against which it is pressed. Wax shall cool and harden quickly, shall not be brittle, and shall be easy to trim with a sharp instrument.

3.2.2 *Vaporization.*—The wax shall leave no solid residue other than carbon when vaporized at 500° C. ± 20° C. (932° F. ± 36° F.).

3.2.3 *Breaking point.*—The wax shall not fracture at a temperature of 23° C. ± 1.1° C. (73.5° F. ± 2° F.) when tested as described in 4.3.3.

3.2.4 *Softening point.* — When tested as specified in 4.3.1, the softening point shall be not less than 60° C. (140° F.) and not more than 70° C. (158° F.).

3.2.5 *Flow.*—When tested as specified in 4.3.2, the maximum flow at 35° C. (95° F.) shall be 10 percent and the minimum flow at 38° C. (100.4° F.) shall be 60 percent.

3.3 Color.

3.3.1 *Class A.*—Class A casting wax shall be pink.

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slightly warm knife or spatula. Provisions shall be made for centering the ball on the upper surface of the specimen. This may be accomplished by the use of a ball guide or by making an indentation in the center of the specimen by pressing the ball or a rounded rod into the material at that point.

4.3.1.3 *Testing procedure.*—The glass container shall be filled to a depth of not less than 3¼ inches with freshly boiled distilled water that has been cooled to 5° C. (41° F.). The ring containing the specimen shall be suspended in the water so that the lower surface of the filled ring is exactly 1 inch above the bottom of the glass container and its upper surface is at least 2 inches below the surface of the water. The ball shall be placed in the water, but not on the specimen. The thermometer shall be suspended so that the bottom of the bulb is level with the bottom of the ring and ¼ inch from the ring. The temperature of the water shall be maintained at 5° C. (41° F.) for 15 minutes, after which time the ball shall be placed in the center of the specimen in the ring. The water shall be stirred throughout the test to maintain a uniform temperature. Heat shall be applied in such a manner that the temperature of the water is raised 5° C. (9° F.) each minute. The rate of rise of temperature shall be uniform and shall not be averaged over the period of the test. The maximum permissible variation for any minute period, after the first three, shall be plus or minus 0.5° C. (0.9° F.). The specimen shall be observed for movement through the ring as the temperature rises. At the instant the specimen touches the bottom of the glass container, the temperature shall be read from the thermometer and the value recorded to the nearest 0.5° C. (0.9° F.) as the softening point of the specimen.

4.3.2 *Flow.*—The mold used in forming the cylindrical flow specimen shall be a flat metal plate 6.0 mm. (0.236 inch) thick, containing one or more holes 10.0 mm. (0.394 inch) in diameter. The mold shall be warmed and placed on a smooth flat surface, which shall also be warmed slightly. The mold and the flat surface upon which it is placed shall be lubricated with a suitable lubricant. Wax that has been heated in a glass beaker over boiling water to just above its melting point, shall be poured into the mold until it is overfilled. As the wax freezes and a shrinkage void or pipe appears, liquid wax shall be added. After cooling, the ends of the specimen shall be carved flush with the surface of the mold. The ends of the specimen shall then be made plane and parallel. The specimen shall be more than 48 hours old and less than 1 week old when tested.

4.3.2.1 *Testing procedure.*—The length of the specimen at room temperature shall be determined with a micrometer and the specimen shall be placed in a water bath and held at testing temperature for 20 minutes prior to testing. The temperature in the water bath shall not vary more than plus or minus 0.1° C. during the tests, and the water shall be stirred throughout the test to maintain a uniform temperature. Thin sheets of cellophane shall be placed between the ends of the specimen and the instrument used for loading. The specimen shall not be removed from the water while the cellophane is being placed. After placing the cellophane, the specimen shall be subjected to a constant axial load of 2000 grams (4.4 pounds) for 10 minutes at testing temperature. The specimen shall then be removed and after it is cooled to room temperature and the cellophane stripped out, the length of the specimen shall be measured at room temperature, and the change in length during the test calculated as a percentage of the original length. The average value of flow at any temperature shall be the average value for three specimens, and shall be reported to the nearest 0.1 percent.

4.3.3 *Breaking point.*—The wax shall be bent double, with the two halves of the specimen in contact along their entire length. The wax shall be bent at a uniform rate over a period of 10 seconds.

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7.2.3.2 *Intermediate packaging.* — Twelve unit cartons (12 units) either shall be packaged in a carton of appropriate size and design, or be completely wrapped with kraft paper of suitable weight. Carton or wrapping shall be adequately secured to prevent accidental opening.

7.2.4 *Packing.*

7.2.4.1 *For domestic shipment.*—Six intermediate packages (72 units) of class A or B, or twelve intermediate packages (144 units) of class C shall be packed in a shipping container conforming to the requirements of Federal Specification NN-B-601, or in a shipping container designed for type 2 load, and conforming to the requirements of Federal Specification NN-B-621 or NN-B-631.

7.2.4.2 *For overseas shipment.*—Six intermediate packages (72 units) of class A or B, or twelve intermediate packages (144 units) of class C shall be packed in a shipping container designed for type 2 load, and conforming to the requirements of Military Specification JAN-P-105, JAN-P-106, or MIL-B-107. Closure and strapping shall conform to the requirements of the appendix to applicable box specification.

7.2.4.2.1 *Case liner.* — Each box shall be lined with a waterproof case liner conforming to the requirements of Military Specification MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto.

7.2.4.3 *Packing variation.*—If the number of units in any shipment is not the exact multiple of the number of units specified to be contained in each shipping container, the units not constituting the nearest such multiple shall be packed in a shipping container of suitable size and design and so constructed as to insure safe delivery to specified destination.

7.2.5 *Marking.*

7.2.5.1 *Packages.* — Each unit carton and each intermediate package shall be marked with stock number, nomenclature, unit, quantity, batch or serial number, office identifying number, and name of contractor.

7.2.5.2 *Shipping containers.* — Each shipping container shall be marked in accordance with Military Standard MIL-STD-129.

7.2.6 *Inspection procedures (Navy only).* —The general inspection procedures shall be in accordance with Navy Department General Specifications for Inspection of Material.

7.2.7 *Ordering data.* — Procurement documents should specify whether the subject commodity is to be packed for domestic or overseas shipment. (See 7.2.4.)

NAVY INTEREST: Md

This specification is a part of Section IV, Part 5, of the Federal Standard Stock Catalog.

Copies of this specification can be purchased for 5 cents each from the General Services Administration, Business Service Center, Region 3, Seventh and D Streets, S. W., Washington 25, D. C. Single copies required for bidding purposes are available without charge at the nearest GSA Regional Office.

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